Collaborative Docs >

Discussion Agenda 06/16/2017 & 07/06/2017 (CGM+Clumps)

Below are discussion agendas for Paper Group-level teleconferences of the AGORA Project. These telecons will focus on planning of the two science-oriented papers to launch during the 2017 summer Workshop: Paper "Clumps" (on June 16, 9 am PDT/6 pm CEST) and Paper "CGM" (on July 6, 11 am PDT/8 pm CEST).

Contents

Information below may have become outdated now. Visit the respective Workspace linked below for updated information.

(last revision: 07/21/2017)

1. Science Papers to Launch in the 2017 Workshop

(1) Paper "Clumps"

--> Below are outdated; visit the Paper "Clumps" Workspace for updated information (comment on July 21, 2017)

	Paper "Clumps"
GOALS	• Compare the off-center star-forming clumps in simulated star-forming galaxies.
LEADERS	• Daniel Ceverino (U. Heidelberg) • Alessandro Lupi (IAP)
PARTICIPANTS	• Interested code representatives: <table><tr><th>Code</th><th>Contacts</th></tr><tr><td>ART-I</td><td>Ceverino, Mandelker, Primack</td></tr><tr><td>GADGET</td><td>Nagamine</td></tr><tr><td>GASOLINE</td><td>Mayer, Wadsley</td></tr><tr><td>GIZMO</td><td>Lupi</td></tr><tr><td>RAMSES</td><td>Agertz</td></tr><tr><td>[to be added]</td><td></td></tr></table>(ICs help: Buehlmann, Hahn / Analysis help: Chakrabarti, Kim, Turk) • Working Groups engaged: - IV (common analysis) - XI (High-redshift Galaxies)
SCIENCE CASE	• Observationally, most star-forming galaxies have off-center clumps (e.g., Guo et al. 2015). • Can we resolve recent conflicting results by ART-I, GASOLINE/ChaNGa, GIZMO? • Numerical issues such as viscosity, dissipation, non-conservation of angular momentum could be discussed. The topic is still impacted by numerical implementation of the hydro as much as subgrid physics. • A related topic is secular disk instabilities. How do different hydro codes (and their gravity solvers) generate bulges of spiral galaxies and impact galactic morphology? • What do we learn from Mayer's latest paper and Mayer/Lupi/Wadsley's earlier paper? Is the difference driven mostly by differences in feedback or in hydro schemes?
	• Pre-production discussion/to-do's: <table><tr><th>Item</th><th>To-do</th></tr></table>

PRE-PRODUCTION DISCUSSION IN THE FIRST MEETING (04/24/2017) AND THE SECOND MEETING (06/16/2017)	Initial condition	• AGORA quiescent 10^{13} Ms halo at z=0. (More information about the halo at **this link**.) It will produce ~10^{12} Ms halo at z=2. • The convex hull ICs found in **Fiacconi et al. 2016** or **Fiacconi et al. 2017** need to be modified for a target redshift of 2 (Fiacconi et al. ICs were for target redshifts of 0 and 6, respectively). Lupi will ask Fiacconi to see if he could remake the IC. Or, Lupi will do it for himself. Either way, by the August Workshop. • In the meantime, a **MUSIC configuration file** for an ellipsoidal IC with levelmax=12 and a Lagrangian region of $4R_{vir}$ at a target redshift 2 is created by Buehlmann and Hahn (**z=2 snapshot of a DM-only test with levelmax=11**, **comparison of IC regions**). • In fact, they now have built a full library of AGORA ICs with different choices of Lagrangian regions for progenitors at different redshifts (to be formally announced soon; **example plot**).
	Target redshift	• z=2
	Definition of a "clump"	• To simplify our comparison, we focus only on the “giant" clumps in a high-z, high-f_{gas} galaxy (i.e., each clump of mass more than 1-2% of the total disk mass, or, each clump with SFR more than 10% of the total disk SFR). • By focusing on "giant" clumps, we can prevent the result of our comparison from depending too much on hard-to-calibrate parameters (e.g., pressure floor). For these "giant" clumps, there also exist good observational counterparts (e.g., Guo et al.). • This means that we do not focus on small "2nd-order" clumps, e.g., small clumps along galactic spiral arms in a low-f_{gas} galaxy.
	Common physics (1): Stellar feedback	• Each group's favorite stellar feedback prescription will be first calibrated across codes to produce similar M_*/M_{halo} ratios, and to overcome the overcooling problem. • ART-I (Ceverino et al.) will use kinetic and thermal feedback modes plus radiation pressure. • Our prediction based on e.g., Ceverino's earlier experience or the **Scylla comparison** is that we don't have to worry too much about this calibration step. • We may consider using isolated disks similar to our disk comparison paper (formerly **Paper "4"**) but with massive gas disks (50% gas fraction or higher), to better understand and align the codes and feedback recipes in this "massive gas disk" regime. • We may also consider evolving and comparing the cosmological halo of 10^{13} Ms at an earlier redshift (e.g., z=4). Hopefully, the pre-calibration in the isolated disk will help us to acquire similar stellar masses across codes.
	Common physics (2): AGN feedback	• We expect that AGN feedback will not be needed as long as we compare galaxies at z=2.
	Common physics (3): Pressure floor	• We will simply adopt the prescription used in our disk comparison paper (formerly **Paper "4"**), because the "giant" clumps do not depend on the pressure floor, but mostly just on Toomre Q.
	Common physics (4): Star formation	• We will start from the prescription used in our disk comparison paper (formerly **Paper "4"**), and see if we need to modify it.
	Common physics (5): Cooling & UVB	• We will adopt the Grackle equilibrium cooling used in our disk comparison paper (formerly **Paper "4"**).
	Output strategy	• See **this link**.
	Common analysis	• We may begin from common analysis routines in **agora-analysis-script** used for our disk comparison paper (formerly **Paper "4"**) and build upon it. • We will build a gas/stellar clump finder in our common analysis script (e.g., simply identifying the high density peaks in Ceverino's previous studies).

		• We may consider identifying and studying additional structures related to gravitational instability (e.g., inflow on the disk to the galactic center, bulges, spirals, etc.)
	Obs. counterpart	• e.g., Guo et al. 2015

FUTURE TIMELINE 2017-2018	• Before the Aug. 2017 Workshop: make cosmological IC ready • During the Aug. 2017 Workshop: finalize runtime parameters (e.g., SF, Fbck), build an analysis shopping list, build and test quick analysis routines to calibrate SF/Fbck prescriptions • After the Aug. 2017 Workshop - Late 2017: run the simulations • Late 2017 - Early 2018: analyze the simulations • Mid 2018: publication target
OTHER LINKS	• WG XI Workspace (High-z Galaxies)

(2) Paper "CGM"

--> Below are outdated; visit the Paper "CGM" Workspace for updated information (comment on July 21, 2017)

	Paper "CGM"
GOALS	• Compare the CGM absorption lines in simulated star-forming galaxies.
LEADERS	• Cameron Hummels (Caltech) • Sijing Shen (U. Oslo)

• Interested code representatives (PARTICIPANTS):

Code	Contacts
ART-I	Mandelker, Primack, Roca-Fabrega, Strawn
ENZO	Hummels, Kim, O'Shea, Wise
GADGET	Nagamine
GASOLINE	Shen, Wadsley
GIZMO	(Hopkins, Hummels)
[to be added]	

PARTICIPANTS	(ICs help: Buehlmann, Hahn, Simpson / Analysis help: Chakrabarti, Fumagalli, Hummels, Kim, Roca-Fabrega, Smith, Turk) • Working Groups engaged: - III (common cosmo ICs) - IV (common analysis) - V (subgrid physics) - IX (Characteristics of Cosmo. Galaxies) - X (Outflows)
SCIENCE CASE	• We will compare the run with observed absorption lines data from e.g., COS-Halos at low redshift (MW-like halo masses, 1.5 < z < 3.5, webpage, Tumlinson et al. 2013), or KBSS at high redshift (Rudie et al. 2012, Steidel et al. 2014). Observers will certainly be interested. • We may later add MHD for comparison.

• Pre-production discussion/to-do's:

Item	To-do
Initial condition	• AGORA quiescent 10^{12} Ms halo at z=0. (More information about the halo at **this link**.) • A **MUSIC configuration file** for an ellipsoidal IC with levelmax=12 and a Lagrangian region of $4R_{vir}$ is created by Buehlmann and Hahn (**z=0 snapshot of a DM-only test with levelmax=11**, **movie**). • In fact, they now have built a full library of AGORA ICs with different choices of Lagrangian regions for progenitors at different redshifts (to be formally announced soon; **example plot**).

	& Resolution	• This IC is designed to give particle masses $M_{DM} = 2.8*10^5$ Ms and $M_{gas} = 6.1*10^4$ Ms, close to the resolution adopted in our disk comparison paper (formerly **Paper "4"**; $M_{gas} = 8.59*10^4$ Ms, dx = 80 pc). We assume that the disk comparison paper found a reasonable way to match simulation resolutions across codes (SPH vs AMR, PLM vs PPM), and will simply piggyback on it.
	Target redshift	• z=3 (initial target), then • z=0 (secondary target)
PRE-PRODUCTION DISCUSSION IN THE FIRST MEETING (04/24/2017) AND THE SECOND MEETING (07/06/2017)	Common physics (1): Stellar feedback	• We will run simulations with two modes of feedback: – (1) "Common" stellar feedback prescription that is identical to our disk comparison paper (formerly **Paper "4"**). This is thermal-only feedback. – (2) "Favorite" stellar feedback prescription(s) of each code group. For "favorite" prescriptions, we ask to provide *at least one* simulation with a mainstream prescription (i.e., thermal+kinetic feedback). Beyond that, you are free to also provide whatever extreme you want to do (e.g., MHD, CR). • The "favorite" feedback prescriptions will need to be first calibrated across codes to produce similar properties (e.g., stellar masses), and to eject metals into CGM. For this, we will carry out a 2-step calibration as follows: – (1) Step #1: We will first run the isolated disk IC (identical one from our disk comparison paper) with both "common" and "favorite" feedback recipes. Each group will tweak the "favorite" recipe(s) to give SFR of 3 Ms/yr at 1 Gyr (see Figure 26 of our disk comparison paper, formerly **Paper "4"**). For this step, you may need to talk to the person who ran the simulation for the disk comparison paper, to acquire the IC, and the parameter set we settled on only after multiple trials and errors. We ask you to carry out this step *before* the August Workshop (see the Future Timeline below). – (2) Step #2: We will then run the cosmological IC of the 10^{12} Ms halo, and compare the stellar masses across codes at z=4. Hopefully, the pre-calibrated "favorite" feedback recipes will give us similar stellar masses.
	Common physics (2): Pressure floor	• We will adopt the prescription used in our disk comparison paper (formerly **Paper "4"**).
	Common physics (3): Star formation	• We will start from the prescription used in our disk comparison paper (formerly **Paper "4"**), and see if we need to modify it.
	Common physics (4): Cooling & UVB	• We will adopt the Grackle equilibrium cooling used in our disk comparison paper (formerly **Paper "4"**).
	Output strategy	• See **this link**.
	Common analysis	• The yt-**Trident** pipeline has been field-tested for a number of codes (ART-I/ENZO/GIZMO), but needs to be fixed for some SPH codes (Cameron; issue reported by Dave/Somerville in GADGET). • Hummels anticipates the new version of yt which enables particle-based codes to be treated more natively (by not depositing to a grid first; see **this link**) to be available in the next few months. We are hopeful that this will be accessible by the August Workshop. • Analysis step could potentially be quite straightforward. Hummels has several scripts that he is willing to share with the community that enable one to do radial profiles (i.e., column density of some ion versus impact parameter) pretty easily, and then we can compare against the COS-Halos results. • One may expect that *all* of the groups are going to be off in trying to match the COS-Halos results at some level, so no one should feel bad about not matching them. It will be instructive to understand why certain groups match some ions and other groups match other ions. • We encourage code groups to submit multiple snapshots with different "favorite" feedback prescriptions. This will produce "brackets" or "bands" in the plots, rather than points. Ideally these "bands" of different codes will overlap with one another, and with the observational data.
	Obs. counterpart	• COS-Halos (see above for references; KBSS are for higher mass halos)
FUTURE TIMELINE		• Before the Aug. 2017 Workshop: calibration step #1 (see above), make cosmological IC ready, learn how to use **Trident** on yt • During the Aug. 2017 Workshop: calibration step #2 (see above), build an analysis shopping list, build and share analysis

2017-2018	scripts • After the Aug. 2017 Workshop - Late 2017: run the simulations, another telecon in October • Late 2017 - Early 2018: analyze the simulations • Spring 2018: paper drafted and submitted
OTHER LINKS	• **Trident download and documentation** • **Trident method paper (Hummels et al.)** • **WG III Workspace (AGORA Cosmological ICs)** • **WG IX Workspace (Characteristics of Cosmological Galaxies)** • **WG X Workspace (Outflows)**

2. Other Science Papers to Consider Later

(1) Paper "Metallicity"

--> Below are outdated; visit the Paper "Metallicity" Workspace for updated information (comment on July 21, 2017)

	Paper "Metallicity"
GOALS	• Connect galaxy evolution and LIGO observations.
LEADERS	(• Sukanya Chakrabarti (Rochester IT) - proposal submitter)
PARTICIPANTS	• Interested code representatives: **Code** / **Contacts** GADGET/GIZMO / Chakrabarti, Lewis GEAR / (Revaz) [to be added] / (Butler may be interested in running tests using an isolated disk IC with ENZO, in connection with Paper "Disk-II"; see Section 3 below.) • Working Groups engaged: - IX (Characteristics of Cosmological Galaxies), - XII (Interstellar Medium)
SCIENCE CASE	• Belczynski et al. 2016 recently showed that the LIGO detection of GW150914 could be explained by the merger of massive, low-metallicity binary stars. A question that remains is what kind of galaxies would these progenitors be typically found in? • Initial work on understanding the host galaxies of LIGO events was done by O'Shaughnessy et al. 2016, showing that dwarf galaxies may contribute significantly to the LIGO signal, due to their low metallicity. Sufficiently low metallicity environments may also be found in the outer disks of spiral galaxies, which have yet to be studied. • The AGORA infrastructure is ideal for testing the effects of feedback on metallicity. It is already clear from Kim et al. 2016 that various solvers have some effect on the metallicity. Different kinds of feedback models are also likely to affect the metallicity in galaxy disks, and given the strong dependence on metallicity on LIGO detections, this in turn is likely to determine the most probable host galaxies for BH-BH mergers. • Regarding connecting LIGO to galaxy evolution, there's a possible connection using chemical evolution within galaxies (e.g., Cote et al. 2017). Interesting opportunities may

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<tr><td></td><td>exist to connect this to AGORA, given that some of the most interesting questions about NS-NS mergers come from the behaviors of ultra-faint dwarf galaxies (e.g., the recent Reticulum II work by Ji et al. 2016 and others). [thoughts by O'Shea]
• Metal diffusion : comparison between different codes/methods [thoughts by Revaz]</td></tr>
<tr><td>PROGRESS MADE OR BEING MADE</td><td>• Pre-production discussion/to-do's:
<table>
<tr><th>Item</th><th>To-do</th></tr>
<tr><td>Initial condition</td><td>• 10^{12} Ms halo at z=0
• We will piggyback on the planned CGM run above and evolve them further down to z=0. In other word, this paper will be one of the causes to push the CGM run to lower redshift.</td></tr>
<tr><td>Common physics</td><td>• Same as the planned CGM run above.</td></tr>
<tr><td>Target redshift</td><td>• z=0</td></tr>
<tr><td>Output strategy</td><td>• See this link.</td></tr>
<tr><td>Common analysis</td><td>• We may begin from metallicity analysis routines in agora-analysis-script used for our disk comparison paper (formerly Paper "4") and build upon it.</td></tr>
<tr><td>Obs. counterpart</td><td>• e.g., Advanced LIGO constraints/predictions in the local Universe (to be determined)</td></tr>
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• Chakrabarti et al. have started a discussion, and to plan test problems using isolated ICs.</td></tr>
<tr><td>OTHER LINKS</td><td>• WG IX Workspace (Characteristics of Cosmological Galaxies)
• WG XII Workspace (Interstellar Medium)</td></tr>
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